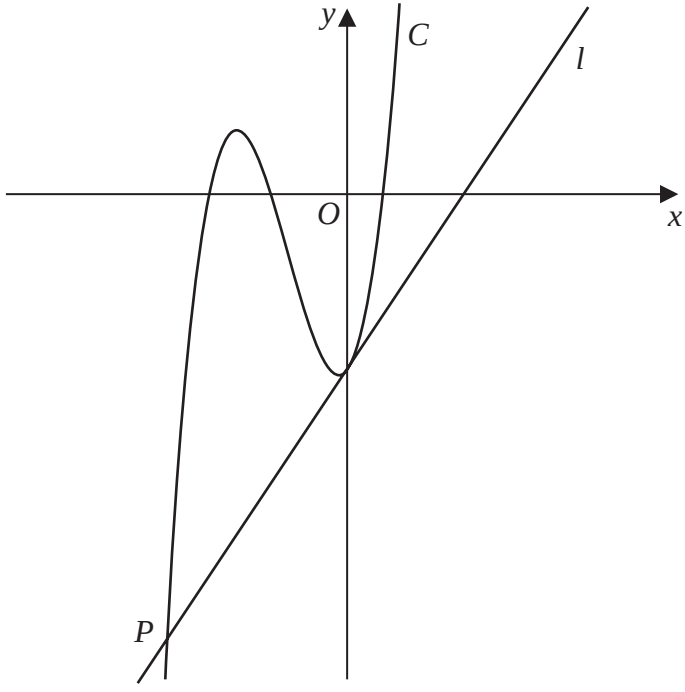



**Figure 4**

Figure 4 shows a sketch of part of the curve $C$ with equation $y = \mathrm{f}(x)$, where

$$\mathrm{f}(x) = (3x + 20)(x + 6)(2x - 3)$$

(a) Use the given information to state the values of $x$ for which

$$\mathrm{f}(x) > 0$$

**(2)**

(b) Expand $(3x + 20)(x + 6)(2x - 3)$, writing your answer as a polynomial in simplest form.

**(3)**

The straight line $l$ is the tangent to $C$ at the point where $C$ cuts the $y$-axis.

Given that $l$ cuts $C$ at the point $P$, as shown in Figure 4,

(c) find, using algebra, the $x$ coordinate of $P$

(*Solutions based on calculator technology are not acceptable.*)

**(5)**